

Corrosion Resistant Poly(cannabinoid) Coating

Caroline Bodnar & Maitreyi Dixit

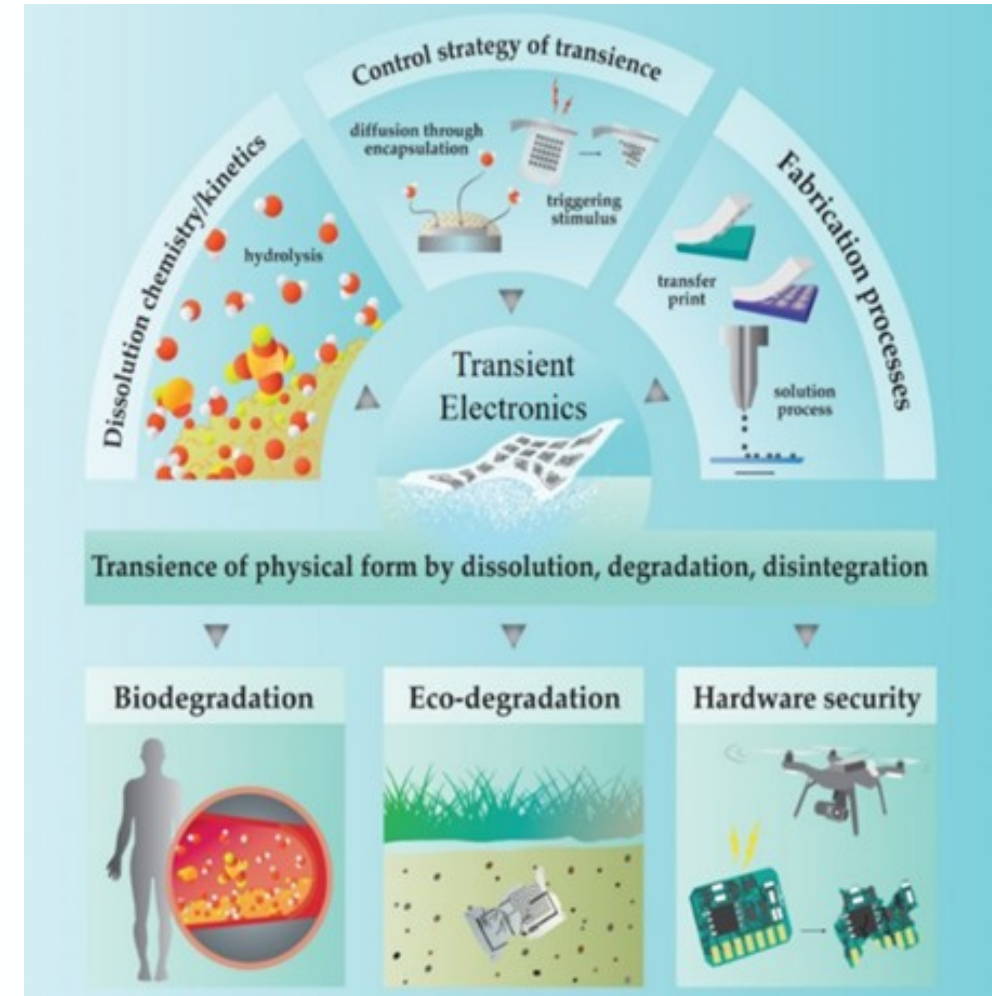
11/28/2023

Overview of the Presentation

- Motivation
- Introduction
- Preliminary Status
- Experimental Setup
 - Arduino
 - Setup Iterations
 - Testing/Data Collection
- Next Steps
- Conclusion and Future Work
- References

Motivation

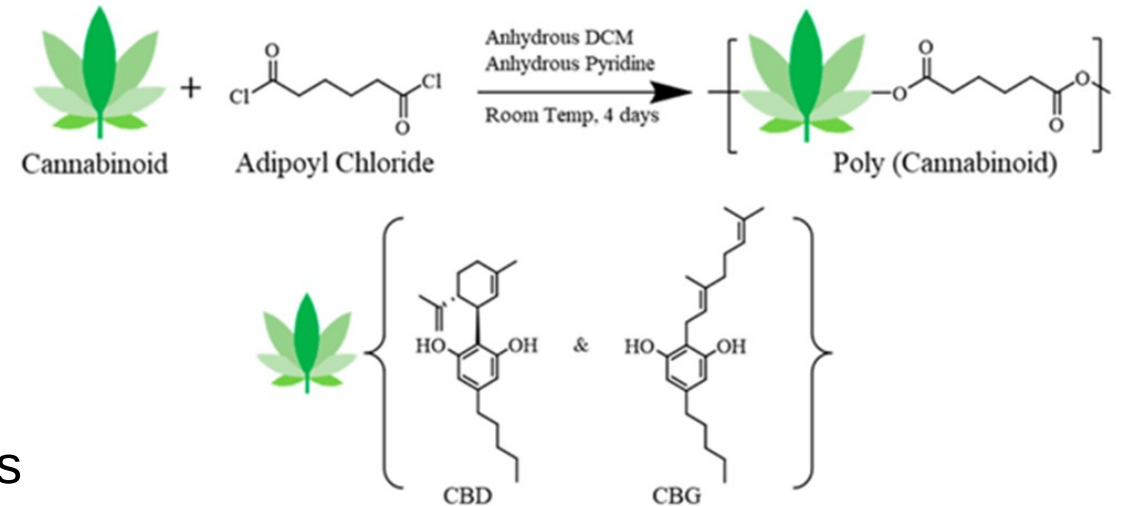
- Increase in electronic waste and pollution → new polymer coating options to improve the sustainability of plastic/electronic industries
- Commonly used polymers include polyanhydrides, polylactic acid, gelatin, polyamines, etc.
 - Weaknesses like storage, weak barrier properties, and requiring additives to degrade
- Transient electronics can apply to applications like biomedical, environmental, and security.



Transient Electronics Applications
[3]

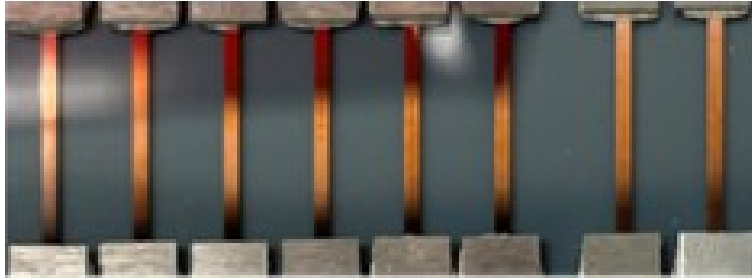
Introduction

- Poly(cannabinoid)s (pCBDs) are an emerging class of biopolymers derived from hemp.
 - Can be used as the coating for transient electronics!
 - Can degrade without additives
 - Hydrophobic
 - Naturally antioxidative
- Research shows that this material can outperform polymers that are generally used as barrier coatings on devices (like PLLA) in terms of antioxidant activities. [4]
- PCBDs are also biodegradable, renewable and more sustainable than other polymers
- Our goal is to investigate and compare these properties to other degradable biopolymers- starting with anti-corrosivity (in different environments)

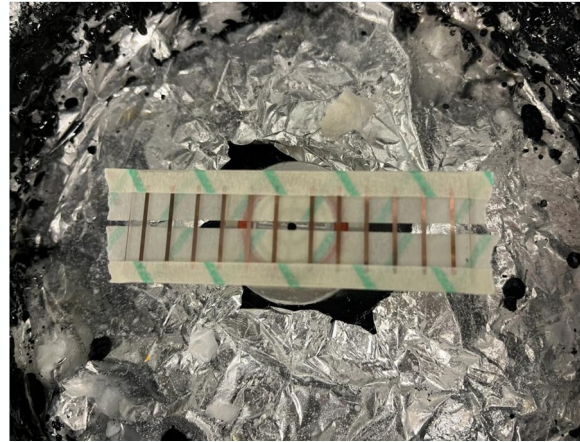


Synthesis of Poly(CBD-Adipate) and Poly(CBG – Adipate)

Methodology

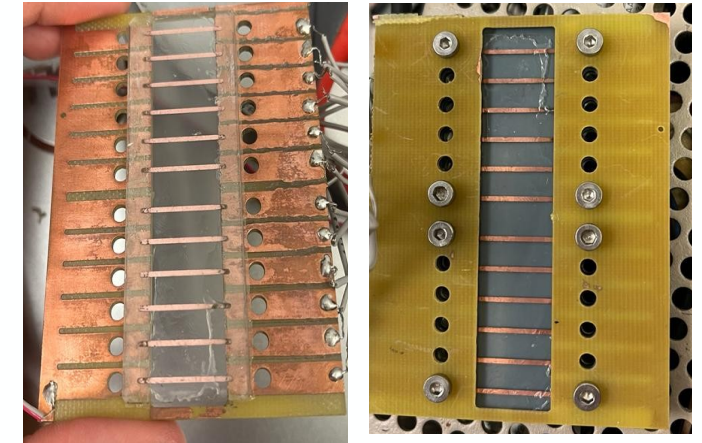


PVD Copper Strips stored in argon



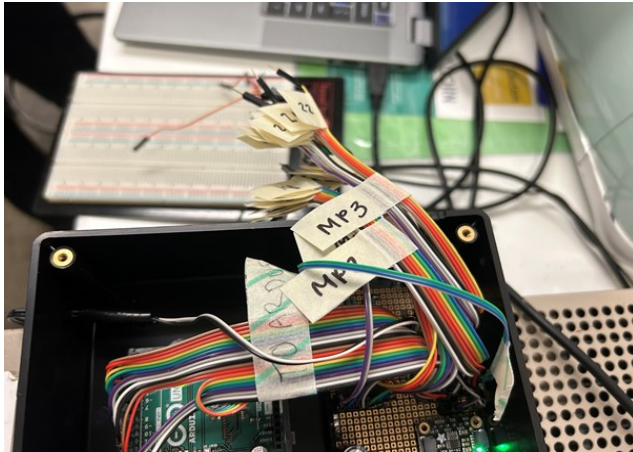
Spin coat PCL-Toluene (1:3 ratio) composition onto PVD copper at about 2000 rpm

Cure at
~100 C

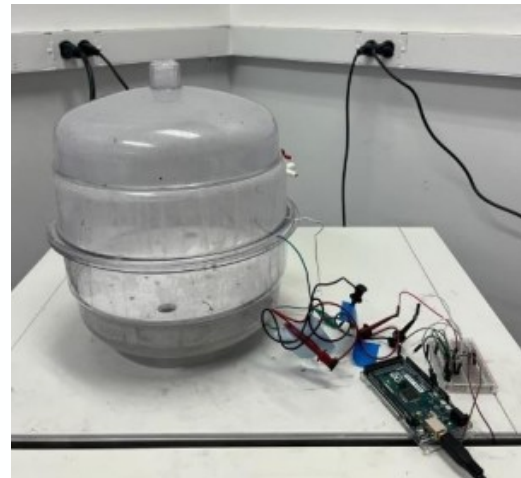


Attach coated PVD glass slide to milled copper plate and clamp down

Cure over
light



Connect copper strip wires to Arduino to read resistance over time



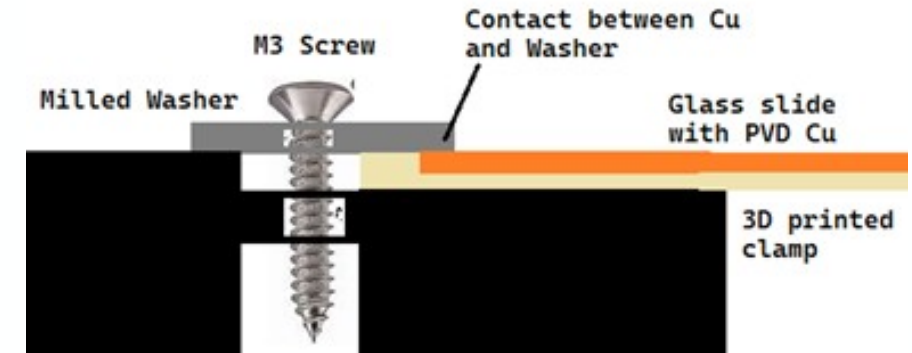
Place in Flower of Sulfur Chamber to corrode copper and facilitate resistance change



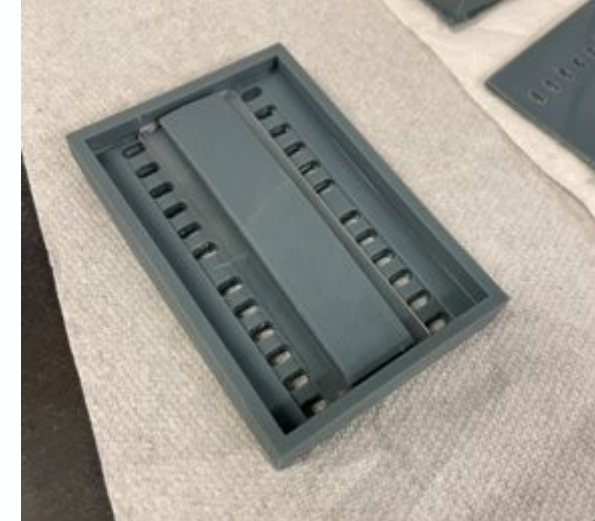
Corroded PVD Copper Strips

Preliminary Status

- Initial methodology proposed with 3D printed tray and washers for the contact
- To be encased in some polymer (separate from coating polymer) to reduce corrosion at contact point
- Physical setup for up to 12 samples at a time
- Arduino code for 1 PVD copper strip/ sample at a time-> with a proposed plan to use multiplexers to increase number of samples read at a time
- Spin coating compositions with PCL and Toluene for a low but still consistent thickness



Proposed Method to clamp PVD samples

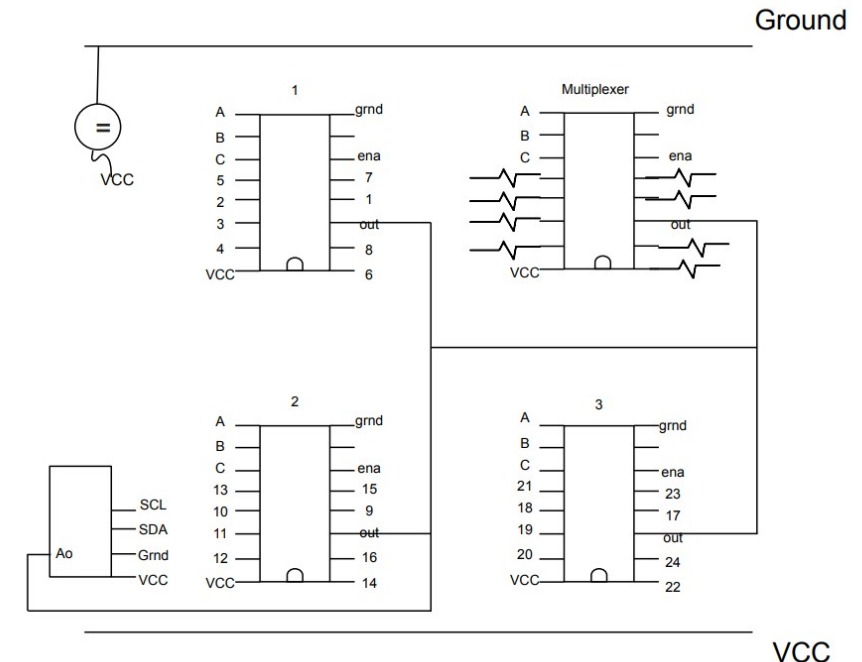
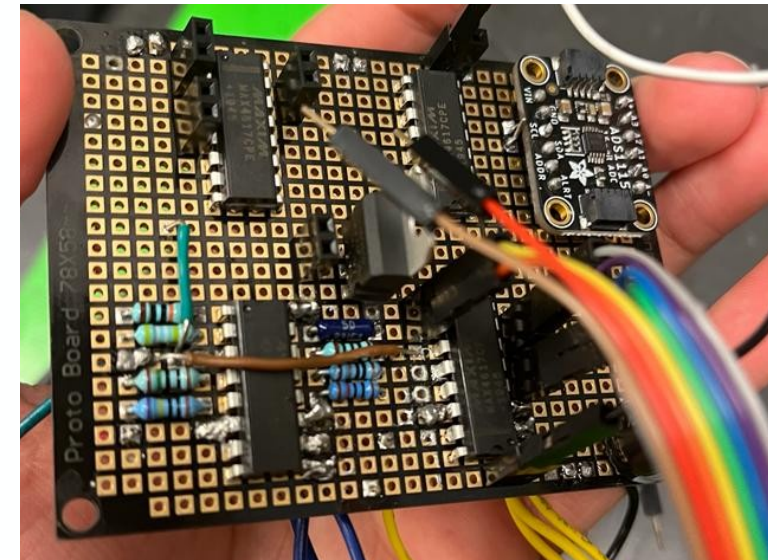


3D printed tray

Experimental Setup

Arduino + Circuit

- Modifying the code and circuit setup for 24 samples at a time, measuring every second
- Using 4 multiplexers- 1 for reference resistors, 3 for samples
- Added an Analog-To-Digital Converter so that Arduino pins could be used to measure voltage drop for all 24 samples
- To streamline setup,
 - created the circuit board
 - removed excess material in the tray
 - included a strip of washers
 - silver conductive paste for contact (instead of encasing)



Circuit Diagram and Circuit Board
with Multiplexer setup

3/31/23

<#>

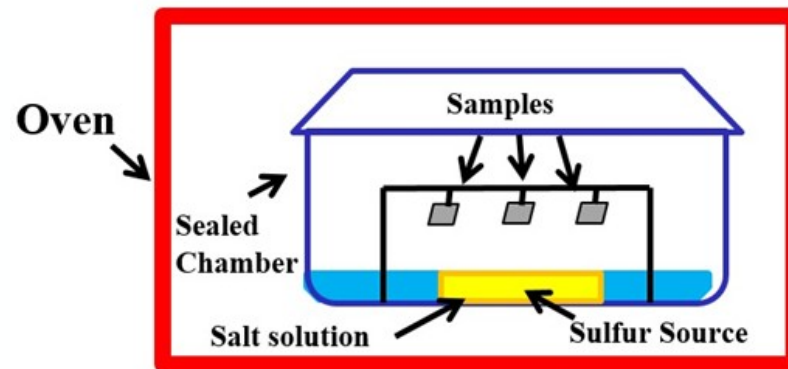
Experimental Setup

Iteration #1 - Fan

- Copper test strips not corroded evenly in flower of sulfur chamber
- Added small fan to the top of the chamber to better circulate vapor and more evenly corrode test copper strips

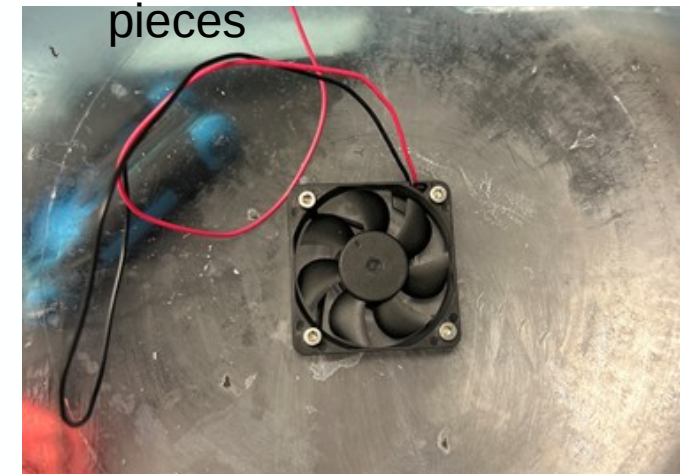


Uneven corrosion across test pieces



Flower of Sulfur Testing (FOS)

FOS process [1]

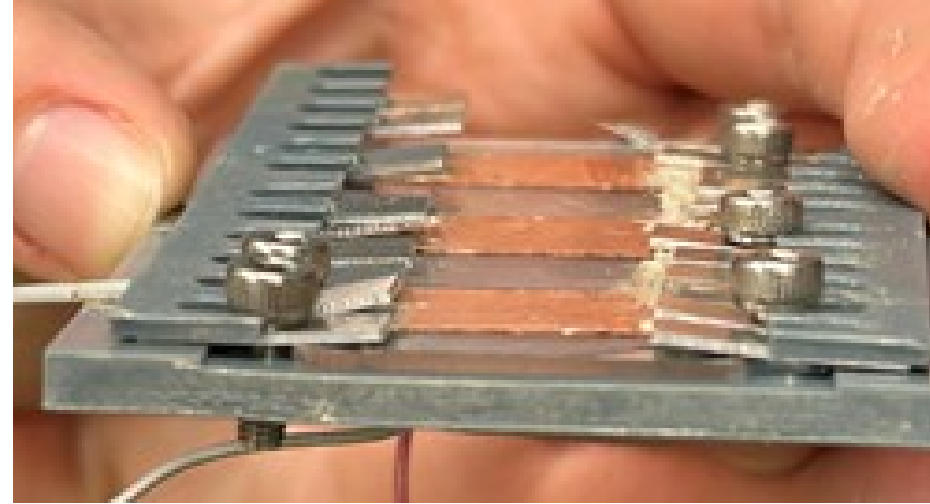


Fan installation

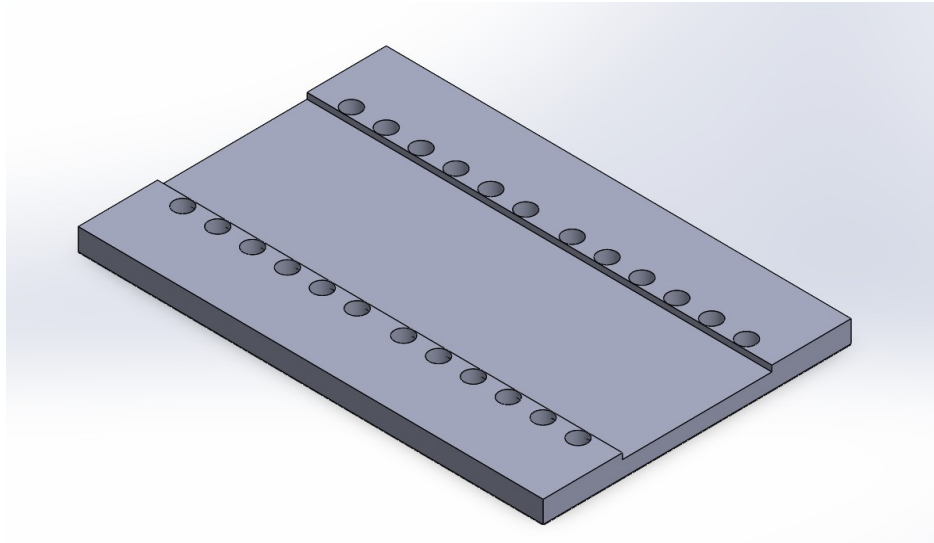
Experimental Setup

Iteration #2 – 3D printed trays

- Testing setup trays made by resin 3D printing
- Experienced bends and cracks when tightening the hardware on the setup
- Increased the thickness of the trays to decrease gaps with glass and better withstand testing



Trays bending



SolidWorks sketch

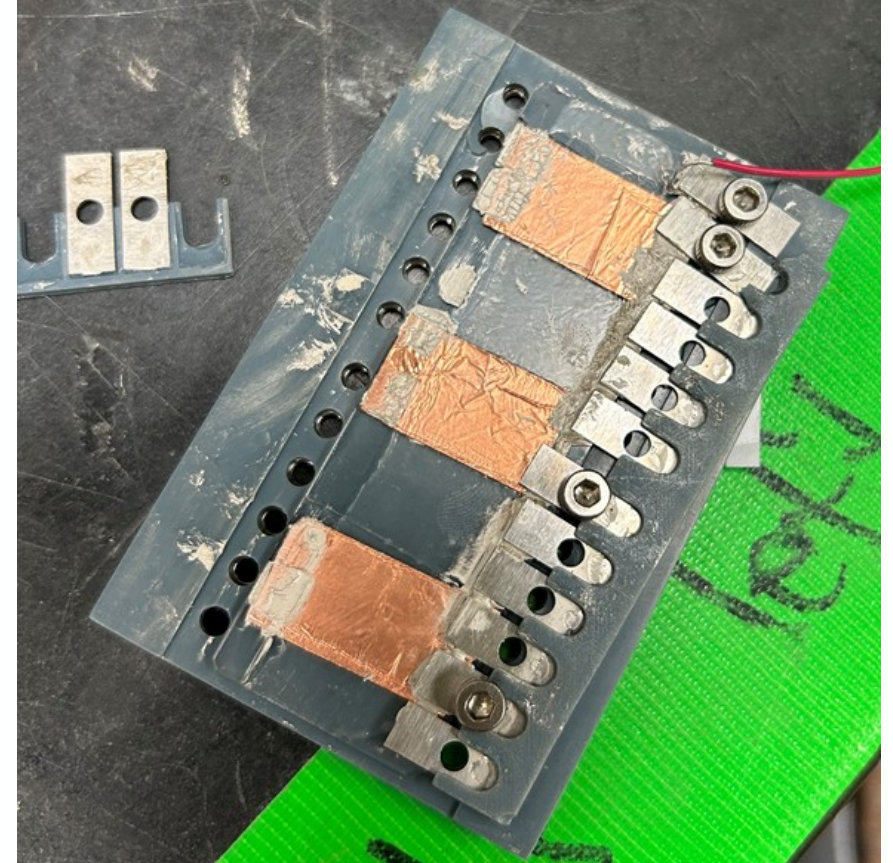


New tray

Experimental Setup

Iteration #3 – New setup

- Silver paste to create conductive connection was messy, time consuming, and inconsistent
- Washers began to break by the time they were tightened enough to secure paste and wires
 - Soldering wires to connect to trays
 - Aluminum washers don't attract solder
 - Attaching wires using ring crimps
 - Ring crimp radius too large



Setup using silver paste and washers



Soldering Method

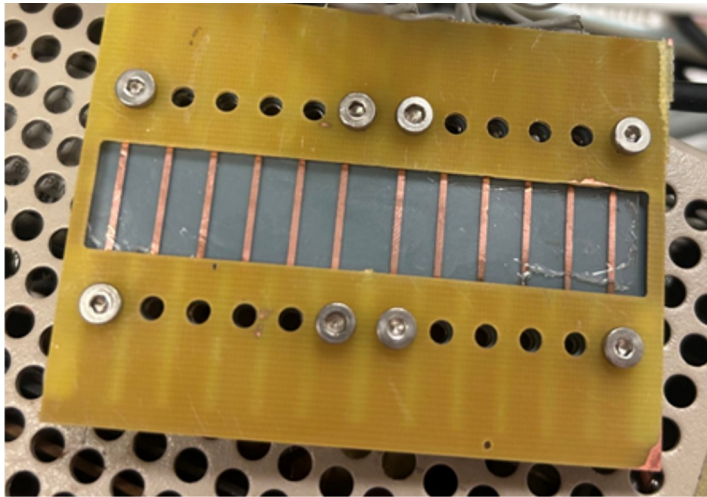


Ring crimp Method

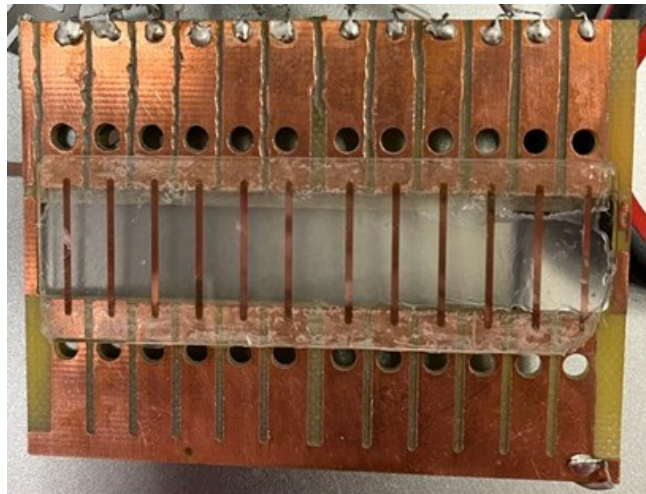
Experimental Setup

Iteration #3 – New setup (cont.)

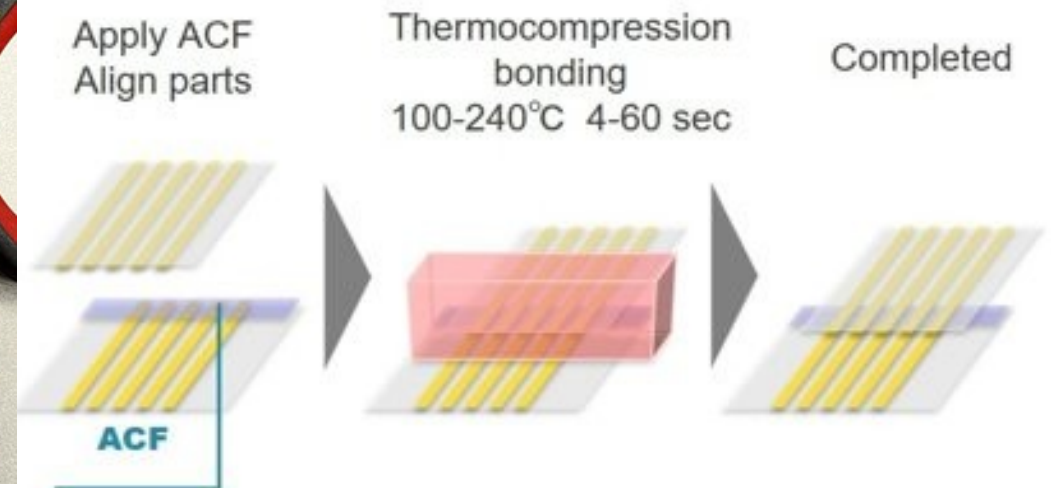
- Created new testing set up by milling laminated PCB copper board
 - Eliminates risk of washers bending and breaking (simplified)
 - Allows wires to be permanently connected using solder
- Replaced silver paste with Anisotropic Conductive Film (ACF) to maintain conductive connection
 - Easier, cleaner, and quicker to apply



Top side of setup



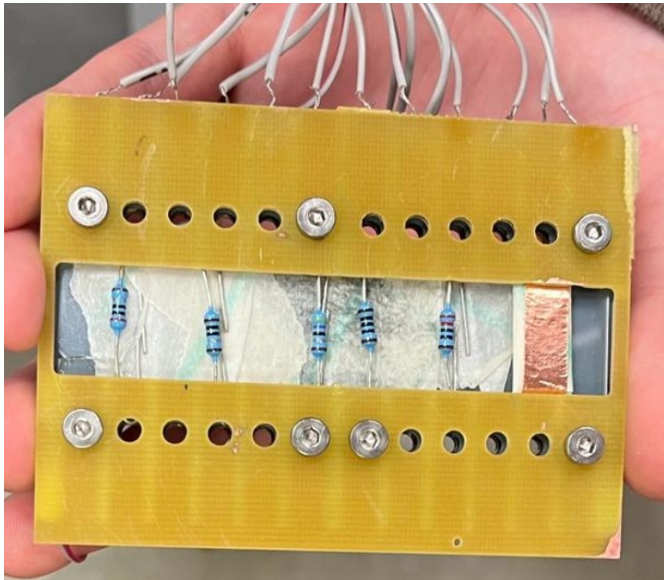
Back side of setup



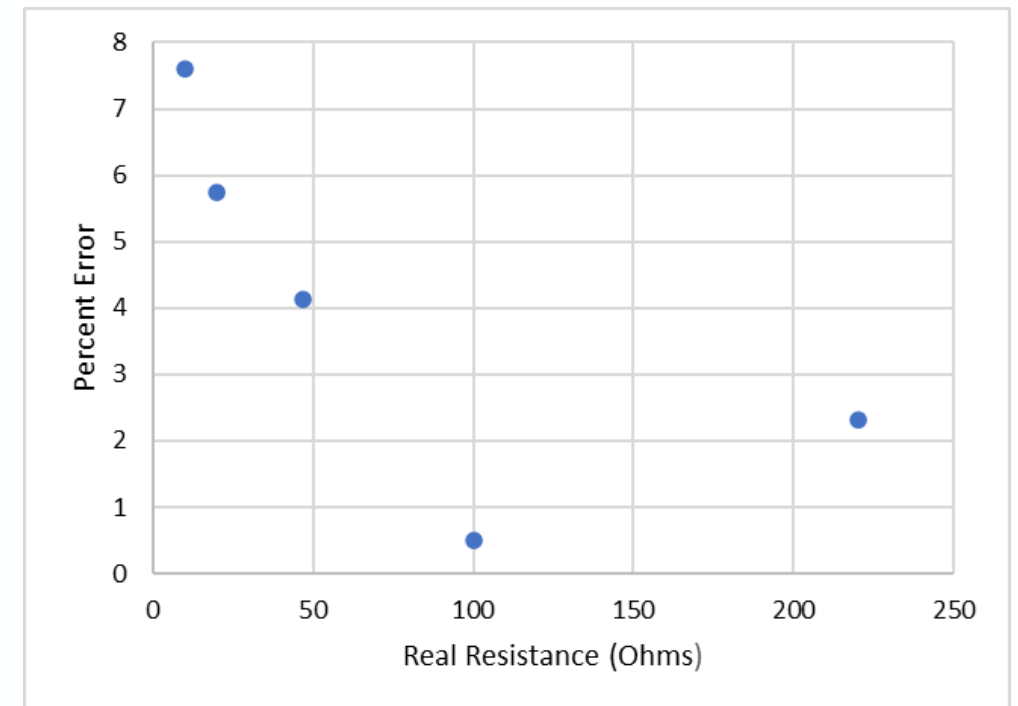
ACF application process
[2]

Testing and Data Collection

- Error testing with resistance values: 10, 20, 47, 100 and 220 Ohms was done
- In general, as predicted, percent error decreased with increase in real resistance values.
- Magnitude of error stayed at around 1 Ohm for all resistors



Real Resistor Value (Ohms)	Magnitude of Error (Ohms)
20	1.15
10	0.76
47	1.94
100	0.51
220	5.1



Next Steps

- Run a full experiment to measure the resistance of the copper over time to gather quantitative data to characterize PCL's effectiveness as a barrier
 - Due to our lack of knowledge and experience with ACFs, potentially return to using silver conductive paste while maintaining simplified milled copper setup
- Eventually compare to the properties of the polycannabinoid



Troubleshooting ACF conductivity by applying thermocompression bonding

Conclusion and Future Work

Conclusions:

- Accuracy of current procedure (results of error testing)
- With a complete, methodized testing setup, we can continue to characterize the protective/barrier properties of PCL, pCBD and compare them to other polymers
- More work is needed to determine the viability of pCBD for implantable and degradable electronics, anti-corrosion coatings, etc.

Further Research:

- Utilizing FTIR to analyze corrosion progress
- Investigate and compare the effects of antioxidant additives (6PPD, curcumin, quercetin)
- Formulation of conductive pCBD inks/pastes

References

- [1] M. Osterman, Flower of Sulfur (FoS) Testing, *A. James Clark School of Engineering*, **2023**
- [2] G. M. Nghiem, K. E. Aasmundtveit, H. Kristiansen and M. Bazilchuk, "Anisotropic Conductive Film (ACF) bonding: effect of interfaces on contact resistance," 2018 7th Electronic System-Integration Technology Conference (ESTC), Dresden, Germany, **2018**, pp. 1-5
- [3] J. Shim, J. A. Rogers, S. Kang, Physically Transient Electronic Materials and Devices, *Materials Science and Engineering: R: Reports*, 2021, 145, 100624
- [4] Daniels, R. (2022). Poly(cannabinoid)s: Hemp-Derived Biocompatible Thermoplastic Polyesters with Inherent Antioxidant Properties. *ACS Applied Materials and Interfaces*.